



LANGUAGE IMPLEMENTERS • CONTRIBUTORS

LANGUAGE CORE MANUAL

ANTLR4 grammar, ASDL AST, builder, unparser, type system,
linter

Open-source Pine Script evaluation framework



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Language Core

ANTLR4 grammar, ASDL AST, builder, visitors, unparser, type system, linter, and error model for PYNE.

Language Core

The language core is the front half of the evaluate contract: source text becomes a typed tree; the tree can be walked, rewritten, dumped, and unparsed without ever entering the bar loop.

Abstract

PYNE treats Pine Script as a **formal pipeline**, not a host plugin. Lexing and parsing are ANTLR4-driven. The concrete syntax tree is lowered by a hand-written visitor (`PinescriptASTBuilder`) into ASDL-generated dataclasses. Public helpers in `helper.py` mirror Python's `ast` module: `parse`, `unparse`, `dump`, `walk`, `literal_eval`. Static tools (linter, type registry, LSP) consume the same tree. Evaluation is out of scope here — see [Runtime](#).

Hard rule: edit grammar only under `resource/`. Paths under `generated/` are regenerated artifacts and must not be hand-edited.

Conceptual model

```

flowchart TB
    SRC["Source string / file"]
    LEX["PinescriptLexer"]
    LEX --> SRC
    LEX --> TOK
    TOK --> PAR
    PAR --> PT
    PT --> BLD
    BLD --> AST
    AST --> VIS
    VIS --> UNP
    UNP --> LIN
    LIN --> TYP
    TYP --> EVAL
    EVAL --> AST

    + LexerBase INDENT/DEDENT"]
    TOK["CommonTokenStream"]
    PAR["PinescriptParser"]
    PT["Parse tree contexts"]
    BLD["PinescriptASTBuilder"]
    ANN["Comment → annotations"]
    AST["ASDL AST"]

    Script | Expression"]
    VIS["NodeVisitor / Transformer"]
    UNP["NodeUnparser"]
    LIN["PineLinter"]
    TYP["TypeRegistry"]
    EVAL["Evaluator / LSP / API"]

    SRC --> LEX --> TOK --> PAR --> PT --> BLD --> AST
    TOK -->|COMMENT tokens| ANN
    BLD --> ANN
    ANN --> AST
    AST --> VIS
    AST --> UNP
    AST --> LIN
    AST --> TYP
    AST --> EVAL

```



Stage	Role	Primary path
Grammar	Concrete syntax	src/pynescrypt/ast/grammar/antlr4/resource/*.g4
ASDL	Algebraic node shape	src/pynescrypt/ast/grammar/asdl/resource/Pinescript.asdl
Builder	CST → AST	src/pynescrypt/ast/builder.py
Helper	Public parse/unparse API	src/pynescrypt/ast/helper.py
Nodes	Re-export generated types	src/pynescrypt/ast/node.py

Interface surface

Consumers almost always enter through `pynescrypt.ast.helper` :

```
from pynescrypt.ast.helper import parse, unparse, dump, walk, literal_eval

tree = parse('//@version=5\nindicator("x")\nplot(close)')
print(dump(tree, indent=2))
print(unparse(tree))
```

Modes:

mode	Entry rule	Root node
"exec" (default)	start_script	Script
"eval"	start_expression	Expression

Full page inventory:

Page	Topic
Grammar (ANTLR4)	Resource vs generated; regen; v6 triple strings
ASDL schema	Algebraic AST; codegen
Builder	CST visitor → ASDL nodes
Helper API	<code>parse</code> , <code>dump</code> , <code>walk</code> , locations
Unparser	Precedence-aware source regen
Visitor / transformer	Read vs rewrite
Type system	Qualifiers, UDT registry
Linter	Static rules, codes
Error model	<code>SyntaxError</code> , indent, listener

Internals

```
src/pynescrypt/ast/
  helper.py           # public pipeline orchestration
  builder.py          # PinescriptASTBuilder
  unparser.py         # NodeUnparser + Precedence
  visitor.py          # NodeVisitor
  transformer.py       # NodeTransformer
  collector.py        # StatementCollector (annotation pairing)
  linter.py           # PineLinter
  type_system.py      # Type*, TypeRegistry, MethodResolver
  error.py            # SyntaxError / IndentationError
  node.py             # from grammar.asdl.generated import *
  grammar/
    antlr4/
      resource/        # EDIT HERE: .g4 + *Base.py
      generated/       # NEVER HAND-EDIT
      error_listener.py
      tool/generate.py
    asdl/
      resource/        # EDIT HERE: Pinescript.asdl
      generated/       # NEVER HAND-EDIT
      tool/generate.py
```

Thin wrappers `grammar/antlr4/lexer.py`, `parser.py`, `visitor.py` re-export generated classes so application code imports stable paths.

Invariants

1. **Resource is source of truth.** Lexer/parser `.g4` and `Pinescript.asdl` define syntax and node shape; generated Python is a build product (committed so install does not require Java).
2. **Round-trip is a test oracle.** For the supported surface, `unparse(parse(src))` should re-parse and preserve semantics (not byte-identical formatting).
3. **Locations are first-class.** Statements and expressions carry `lineno`, `col_offset`, optional `end_*` for diagnostics and LSP.
4. **Annotations are not free-floating.** Comment tokens on `COMMENT_CHANNEL` are reified as `Comment` nodes and attached to `Script` / `FunctionDef` / `TypeDef` / `Assign` via kind suffixes (`S`, `F`, `T`, `V`).
5. **Recursion budget.** Deep ternary chains raise the Python recursion limit temporarily during parse (cap ≥ 5000).

Worked examples

Minimal script:



```
from pynescrypt.ast.helper import parse, dump

src = """
//@version=5
indicator("demo")
x = ta.sma(close, 14)
plot(x)
"""

tree = parse(src)
assert tree.__class__.__name__ == "Script"
assert any("version" in a for a in (tree.annotations or []))
print(dump(tree, include_attributes=True, indent=2))
```

Expression-only eval mode:

```
from pynescrypt.ast.helper import parse, literal_eval

expr = parse("1 + 2 * 3", mode="eval")
assert literal_eval(expr) == 7
```

Failure modes

Symptom	Likely cause	Where to look
<code>SyntaxError</code> with caret	Lexer/parser reject	Error model
<code>IndentationError</code>	Bad INDENT/DEDENT from LexerBase	<code>PinescriptLexerBase.py</code>
Missing <code>visit_*</code> after grammar change	Builder not updated for new rule	Builder
Hand-edit of <code>generated/</code> lost on regen	Violated resource-only rule	Grammar
Full regen breaks <code>builder.py</code>	Parser context accessors changed	Prefer targeted lexer refresh (v6 case study)

See also

- Grammar (ANTLR4)
- Helper API
- Runtime
- LSP
- Compatibility

Grammar (ANTLR4)

Resource .g4 grammars, generated artifacts, INDENT/DEDENT, regeneration, and the never-edit-generated rule.

Grammar (ANTLR4)

Concrete syntax for Pine Script in PYNE is defined by a split lexer/parser ANTLR4 grammar plus hand-written base classes that inject Python-like indentation tokens.

Abstract

Two grammars live under `src/pynscript/ast/grammar/antlr4/resource/`:

- `PinescriptLexer.g4` — keywords, operators, literals (including v6 triple-quoted strings), comments, hidden whitespace.
- `PinescriptParser.g4` — statements, expressions, type/enum/function/method declarations, control structures.

Python runtime classes under `generated/` are **outputs** of `antlr4 -Dlanguage=Python3`. They are committed so end users never need a JDK. **Never edit files under `generated/`** — changes evaporate on the next regeneration and fight the build pipeline.

Hand-written bases (copied into `generated/` by the generate tool):

- `resource/PinescriptLexerBase.py` — INDENT/DEDENT, newline elision, multiline string handling.
- `resource/PinescriptParserBase.py` — parser superClass hooks.



Conceptual model

```
flowchart LR
    subgraph resource ["resource/ – edit here"]
        LG4["PinescriptLexer.g4"]
        PG4["PinescriptParser.g4"]
        LB["PinescriptLexerBase.py"]
        PB["PinescriptParserBase.py"]
    end

    GEN["antlr4 tool"]
    subgraph generated ["generated/ – do not hand-edit"]
        LPY["PinescriptLexer.py"]
        PPY["PinescriptParser.py"]
        VIS["PinescriptParserVisitor.py"]
        LIS["PinescriptParserListener.py"]
        LBgen["*Base.py copies"]
    end

    WRAP["lexer.py / parser.py / visitor.py wrappers"]
    BLD["builder.py visit_*"]

    LG4 --> GEN
    PG4 --> GEN
    GEN --> LPY
    GEN --> PPY
    GEN --> VIS
    GEN --> LIS
    LB --> LBgen
    PB --> LBgen
    LPY --> WRAP
    PPY --> WRAP
    VIS --> BLD
```

Interface surface

Application code should not import `generated` symbols directly when a wrapper exists:

```
from pynescrypt.ast.grammar.antlr4.lexer import PinescriptLexer
from pynescrypt.ast.grammar.antlr4.parser import PinescriptParser
from pynescrypt.ast.grammar.antlr4.visitor import PinescriptParserVisitor
from pynescrypt.ast.grammar.antlr4.error_listener import PinescriptErrorListener
```

The public parse path is higher level — **Helper API** — which constructs lexer, token stream, parser, and strips default console error listeners in favor of `PinescriptErrorListener`.

Entry rules (`PinescriptParser.g4`)

Rule	Purpose
<code>start</code> → <code>start_script</code>	Full script (<code>statements?</code> EOF)
<code>start_expression</code>	Single expression + optional NEWLINE + EOF
<code>start_comments</code>	Comment stream only



Lexer structural facts

Token / channel	Behavior
INDENT / DEDENT	Imaginary tokens emitted by <code>PinescriptLexerBase</code> (declared in <code>tokens { }</code>)
COMMENT	<code>//</code> line comments → <code>COMMENT_CHANNEL</code> (not the default channel)
WS	Spaces/tabs/form-feed → <code>HIDDEN</code>
ERROR_TOKEN	Catch-all <code>.</code> for recovery/reporting
TRIPLE_DQ_STRING / TRIPLE_SQ_STRING	v6 multiline strings, typed as <code>STRING</code>

Keywords include `var`, `varip`, `method`, `export`, `enum`, `type`, qualifiers `const` / `input` / `simple` / `series`, and control forms `if` / `for` / `while` / `switch`.

Parser structural facts

- **Compound vs simple statements.** Functions, methods, types, enums, and structure-valued assignments are compound; imports, break/continue, expression statements, and simple assignments terminate with `NEWLINE`.
- **Structures are dual-use.** `if` / `for` / `while` / `switch` appear both as statements and as **expressions** (`structure_expression`) — the classic Pine “if returns a value” form.
- **Local blocks.** Either indented (`NEWLINE INDENT statements DEDENT`) or inline single statement.
- **Library export.** `EXPORT?` prefixes function/method/type/enum and June-2025 `export const T name = ...` compound initializers.

Internals

Paths

Path	Edit?	Role
<code>src/pinescript/ast/grammar/antlr4/resource/PinescriptLexer.g4</code>	Yes	Lexer grammar
<code>src/pinescript/ast/grammar/antlr4/resource/PinescriptParser.g4</code>	Yes	Parser grammar
<code>src/pinescript/ast/grammar/antlr4/resource/PinescriptLexerBase.py</code>	Yes	Indentation machine
<code>src/pinescript/ast/grammar/antlr4/resource/PinescriptParserBase.py</code>	Yes	Parser base
<code>src/pinescript/ast/grammar/antlr4/generated/**</code>	No	ANTLR output + copied bases
<code>src/pinescript/ast/grammar/antlr4/tool/generate.py</code>	Tool	Runs <code>antlr4</code> , copies <code>*.py</code> bases
<code>src/pinescript/ast/grammar/antlr4/error_listener.py</code>	Hand	Maps ANTLR errors → <code>ast.error.SyntaxError</code>

LexerBase responsibilities

`PinescriptLexerBase` is not optional sugar; it is part of the language definition:

- Ignore leading / collapse consecutive newlines; ensure trailing newline.

- Suppress newlines inside open `()` / `[]`, after operators, and for soft wrap lines whose indent width is not a multiple of four.
- Push `INDENT` / `DEDENT` with tab length and indent length = 4.
- Special-case multiline string tokens so wrap indentation inside strings is not mis-tokenized as structure.

Indent mistakes surface as `pynescrypt.ast.error.IndentationError` (subclass of the project `SyntaxError`).

Regeneration

```
# Preferred hatch env (pulls antlr4-cli)
hatch run lint:gen-parser

# Or: python -m pynescrypt.ast.grammar.antlr4.tool.generate
# which invokes: $(python)/antlr4 -o generated -lib resource -listener -visitor -Dlanguage=Python3
# then copies resource/*.py into generated/
```

Requires a working `antlr4` CLI (often `antlr4-cli` on `PATH`, sometimes under `~/.local/bin` or next to `sys.executable`). Generated modules are excluded from mypy/ruff strictness in `pyproject.toml` — do not “lint-fix” them by hand.

Downstream after grammar change

1. Regenerate (or selectively refresh lexer — see case study).
2. Add/adjust `visit_*` methods in `src/pynescrypt/ast/builder.py` for new or renamed rules.
3. If the **shape** of the language changed (new statement/expression kind), update `Pinescript.asdl` and regenerate ASDL nodes — [ASDL schema](#).
4. Smoke-test with `parse` / `unparse` on a minimal snippet **before** the full `tests/data/builtin_scripts/corpus`.

Invariants

1. **Never hand-edit** `generated/`. Patches belong in `resource/*.g4` or `resource/*Base.py`.
2. **Visitor contract.** Generated `PinescriptParserVisitor` method names follow ANTLR rule names; the builder subclass must track renames.
3. **Comments are off the default channel.** Annotation logic in `helper._collect_comment_nodes` iterates `token_stream.tokens` for `COMMENT` type after `fill()`.
4. **Indent unit is four spaces.** Soft-wrap lines that are not multiples of four are newline-elided, not nested blocks.

Worked examples

Minimal grammar-level mental model

```
//@version=5
indicator("x")
if close > open
    plot(1)
else
    plot(0)
```

Lexer emits NEWLINE, INDENT, DEDENT around the `if` bodies. Parser builds `if_structure` → builder yields `ast.If` with `body` / `orelse` statement lists.

v6 triple-quoted strings

Resource lexer rules (illustrative — see live `PinescriptLexer.g4`):

```
TRIPLE_DQ_STRING : '"""' ( ~'"""' | '"""' ~'"""' | '"""' ~'"""' )* '"""' -> type(String) ;
TRIPLE_SQ_STRING : '\'\'\'\'\' ( ~'\'' | '\'' ~'\'' | '\'' ~'\'' )* '\''\'\'\' -> type(String) ;
```

ANTLR quoting pitfall: putting `"""` or `'''` as fragment literals can produce tool errors (`quote` came as a complete surprise). Factor starters when needed:

```
fragment TRIPLE_SQ_START: '\'' '\'' '\'';
```

Content and source indentation inside the triple quotes are preserved literally (no automatic dedent).

Case study: targeted lexer refresh (2026-07)

Full regeneration once produced a `PinescriptParser.py` whose context accessors diverged from what the hand-written builder expected (e.g. missing `template_spec_suffix()`), breaking even trivial parses. The practical recovery:

1. Edit **only** `resource/` grammar.
2. Generate the lexer in a clean temp directory (avoids nested `src/` mirror paths).
3. Copy **only** `PinescriptLexer.py` (+ refreshed `LexerBase.py`) into `generated/` .
4. Leave committed `PinescriptParser.py` and visitors untouched unless prepared to patch the builder in the same change.
5. Verify immediately:

```
from pinescript.ast.helper import parse, unparse
ast = parse('indicator("x")\ns = ""\nfoo\n bar\n""\n')
assert "foo" in unparse(ast)
```



Failure modes

Failure	Cause	Mitigation
<code>mismatched input</code> after <code>"""</code>	Old lexer ATN without triple rules	Refresh generated lexer from resource
Full regen breaks builder	Parser context API drift	Selective copy; or update all <code>visit_*</code> together
<code>antlr4</code> not found	CLI not on PATH	Use hatch env or full path to <code>antlr4-cli</code>
Nested <code>generated/src/pynescrypt/...</code> junk	Running antlr with wrong <code>-o</code> cwd	Generate from clean temp + explicit <code>-o</code>
IndentationError mid-file	Mixed tabs/spaces or wrong nest	Align to 4-space blocks; check soft-wrap rules
Silent loss of hand patches	Edited <code>generated/</code>	Revert; re-apply in <code>resource/</code>

See also

- [ASDL schema](#)
- [Builder](#)
- [Error model](#)
- [Helper API](#)
- [Missing features](#)

ASDL Schema

Algebraic AST definition in Pinescript.asdl, generated dataclasses, and how schema changes flow into the pipeline.

ASDL Schema

The abstract syntax tree is not free-form dicts. It is an **algebraic data type** declared in ASDL and materialized as Python dataclasses.

Abstract

`src/pynescrypt/ast/grammar/asdl/resource/Pinescript.asdl` defines the `Pinescript` module: roots (`Script`, `Expression`), statements, expressions (including control structures that are also expressions), operators, params, args, cases, and comments. Codegen writes `grammar/asdl/generated/PinescriptASTNode.py`. `src/pynescrypt/ast/node.py` re-exports that module so the rest of the codebase imports `from pynescrypt.ast import node as ast` (or `from pynescrypt.ast.node import Script`).

ASDL is the contract between **builder** (produces nodes), **evaluator/LSP** (consume nodes), and **unparser** (serializes nodes). Changing a field without regenerating and updating all three is a schema break.

Conceptual model

```

flowchart TB
    ASDL["Pinescript.asdl"]
    resource["resource"]
    GEN["asdl / pyasdl generator"]
    PY["PinescriptASTNode.py"]
    generated_dataclasses["generated dataclasses"]
    NODE["node.py re-export"]
    BLD["builder.py"]
    UNP["unparser.py"]
    VIS["visitor / transformer"]
    EV["evaluator / LSP"]

    ASDL --> GEN
    GEN --> PY
    PY --> NODE
    NODE --> BLD
    NODE --> UNP
    NODE --> VIS
    NODE --> EV
  
```

Each product sum type becomes a Python class hierarchy under `AST` with:

- `_fields` — logical children (walked by `iter_fields` / visitors)
- `_attributes` — location metadata (`lineno`, `col_offset`, `end_lineno`, `end_col_offset`) where declared



Interface surface

Module roots (`mod`)

Constructor	Fields	Use
<code>Script</code>	<code>stmt* body</code> , <code>string* annotations</code>	Full file (<code>mode="exec"</code>)
<code>Expression</code>	<code>expr body</code>	Single expression (<code>mode="eval"</code>)

Statements (`stmt`)

Constructor	Notable fields
<code>FunctionDef</code>	<code>name</code> , <code>param* args</code> , <code>body</code> , <code>method?</code> , <code>export?</code> , <code>annotations</code>
<code>TypeDef</code>	UDT <code>type Name</code> <code>body</code> (fields + methods as stmts)
<code>EnumDef</code>	<code>enum Name</code> <code>members</code>
<code>Assign</code>	<code>target</code> , optional <code>value / type / mode (Var / VarIp)</code> , <code>export?</code> , <code>annotations</code>
<code>ReAssign</code>	<code>:=</code> reassignment
<code>AugAssign</code>	<code>+=</code> ... ops
<code>Import</code>	<code>namespace/name/version</code> , optional <code>alias</code>
<code>Expr</code>	Expression statement
<code>Break / Continue</code>	Loop control

All `stmt` and most other attributed products carry location attributes.

Expressions (`expr`)

Operators and primaries:

- `BoolOp`, `BinOp`, `UnaryOp`, `Conditional ( ? : )`, `Compare`
- `Call`, `Constant`, `Attribute`, `Subscript`, `Name`, `Tuple`
- Structures as expressions:** `ForTo`, `ForIn`, `While`, `If`, `Switch` — same shapes can appear as values (Pine's expression-oriented control flow)
- `Qualify` — `const / input / simple / series` applied to a type/expression
- `Specialize` — generic specialization form (`value + args`)



Auxiliaries

Product	Role
<code>decl_mode</code>	<code>Var</code> <code>VarIp</code>
<code>type_qual</code>	<code>Const</code> <code>Input</code> <code>Simple</code> <code>Series</code>
<code>expr_context</code>	<code>Load</code> <code>Store</code>
<code>param</code>	Function/method parameter
<code>arg</code>	Call argument (optional keyword <code>name</code>)
<code>case</code>	Switch arm (<code>pattern?</code> , <code>body</code>)
<code>cmnt</code> / <code>Comment</code>	Comment with <code>value</code> and <code>kind</code> (annotation classification)

Operators are empty product constructors (`Add` , `And` , `Eq` , ...) used as tags — the unparser maps class names back to tokens.

Importing nodes

```
from pynescrypt.ast import node as ast

script = ast.Script(body=[
    ast.Expr(value=ast.Call(
        func=ast.Name(id="plot", ctx=ast.Load()),
        args=[ast.Arg(value=ast.Name(id="close", ctx=ast.Load()))],
    ))
])
```

Prefer constructing via `parse()` unless synthesizing trees for tests.

Internals

Paths

Path	Edit?
<code>src/pynescrypt/ast/grammar/asdl/resource/Pinescript.asdl</code>	Yes — schema source of truth
<code>src/pynescrypt/ast/grammar/asdl/generated/PinescriptASTNode.py</code>	No — regenerated
<code>src/pynescrypt/ast/grammar/asdl/tool/generate.py</code>	Generator entry
<code>src/pynescrypt/ast/node.py</code>	Thin re-export (<code>from .grammar.asdl.generated import *</code>)

Generated class shape

Excerpt of generated form (do not edit this file; shown for orientation):



```
@_dataclasses.dataclass
class Script(mod):
    body: list[stmt] = field(default_factory=list)
    annotations: list[string] = field(default_factory=list)
    _fields: ClassVar[list[str]] = ["body", "annotations"]
```

`stmt` base injects location `_attributes`. Hashability is forced to `object.__hash__` so nodes can live in sets/maps by identity when needed.

Regeneration

When the schema changes:

```
# Project convention (pyasdl / asdl tool – see asdl/tool/generate.py)
pyasdl src/pynescrypt/ast/grammar/asdl/resource/Pinescript.asdl \
  -o src/pynescrypt/ast/grammar/asdl/generated
```

Then update, in the same change set:

1. `builder.py` construction sites
2. `unparser.py` `visit_*` methods
3. Evaluator expression/statement handlers
4. Any isinstance checks in LSP features

Dual nature of structures

ASDL places `If`, `ForTo`, `ForIn`, `While`, `Switch` under `expr`, while the builder often wraps structure-as-statement under `Expr(value=...)`. Annotation collection (`StatementCollector`) treats `assign/reassign/augassign/expr` that hold structure values specially so nested statements remain visible for `//@...` pairing.

Invariants

1. **Schema first.** New language constructs need an ASDL constructor (or a clear encoding into an existing one) before evaluator semantics.
2. **Fields vs attributes.** Visitors walk `_fields` only. Location is metadata; missing locations are fillable via `fix_missing_locations`.
3. **Optional flags as `int?`.** `method` and `export` are integer flags (truthy when set), not booleans — match builder assignments (`export = 1`).
4. **Generated file is not a style playground.** Formatting/lint of `PinescriptASTNode.py` is overwritten on regen.

Worked examples

Map ASDL to a short script

```
//@version=5
f(x) => x + 1
a = f(close)
```



Approximate tree:

```
Script(  
  annotations=["//@version=5", ...],  
  body=[  
    FunctionDef(name="f", args=[Param(name="x")], body=[Expr(BinOp(...))]),  
    Assign(target=Name("a", Store), value=Call(...)),  
  ],  
)
```

Dump fields programmatically

```
from pynescrypt.ast.helper import parse, iter_fields  
  
tree = parse("x = 1")  
for name, value in iter_fields(tree):  
    print(name, type(value).__name__)
```

Failure modes

Failure	Cause
<code>AttributeError</code> on new field	Schema regenerated but consumer not updated
Builder constructs wrong product	ASDL and grammar out of sync
Unparser omits new node	Missing <code>visit_NewNode</code> → <code>generic_visit</code> may emit nothing useful
Hand-edit of generated nodes lost	Edited <code>generated/</code> instead of <code>.asdl</code>

See also

- [Builder](#)
- [Visitor / transformer](#)
- [Unparser](#)
- [Grammar \(ANTLR4\)](#)

AST Builder

PinescriptASTBuilder: ANTLR parse-tree visitor that constructs ASDL nodes, locations, and comment kinds.

AST Builder

The builder is the semantic boundary between ANTLR's concrete parse tree and PYNE's ASDL abstract tree.

Abstract

`src/pinescript/ast/builder.py` defines three cooperating mixins:

1. **PinescriptASTLocator** — maps `ParserRuleContext.start/stop` tokens to `lineno` / `col_offset` / `end_*`.
2. **PinescriptCommentParser** — classifies `//...` text into annotation kinds (`@=S`, `@0F`, `#`, plain `//`, ...).
3. **PinescriptASTBuilder** — subclasses generated `PinescriptParserVisitor`, implements `visit_*` for every meaningful rule, returns ASDL nodes.

`helper._parse` constructs a builder, runs `builder.visit(rule_context)`, then (in exec mode) uses `StatementCollector` + comment tokens to attach annotations. The builder itself does not walk the default token channel for comments; that is a post-pass.

Conceptual model

```
sequenceDiagram
    participant H as helper._parse
    participant L as PinescriptLexer
    participant P as PinescriptParser
    participant B as PinescriptASTBuilder
    participant C as StatementCollector

    H->>L: tokenize stream
    H->>P: start_script / start_expression
    P-->>H: parse tree root ctx
    H->>B: visit(root)
    B-->>H: Script | Expression
    H->>C: visit(Script) -> stmts
    H->>B: _parseComment on COMMENT tokens
    H->>H: _add_annotations(script, stmts, comments)
```

Interface surface

You rarely instantiate the builder directly; `parse()` does. For tools that already hold a parse tree:

```
from pinescript.ast.builder import PinescriptASTBuilder
from pinescript.ast.grammar.antlr4.parser import PinescriptParser
# ... construct parser, parse to ctx ...
node = PinescriptASTBuilder().visit(ctx)
```



Locator API

Method	Role
<code>_getLocations(ctx)</code>	Dict of four location keys
<code>_setLocations(node, ctx)</code>	Mutates node attributes from <code>ctx.start</code> / <code>ctx.stop</code>

End column calculation accounts for embedded newlines in the stop token text.

Comment kinds

`_parseComment` returns `(kind, parts)`:

Pattern	Kind prefix	Suffix	Example
<code>//@version = 5</code>	<code>@=</code>	S (script)	version assignment
<code>//@description ...</code>	<code>@0</code>	S	script description
<code>//@function / @returns</code>	<code>@0</code>	F	function docs
<code>//@type</code>	<code>@0</code>	T	type docs
<code>//@variable</code>	<code>@0</code>	V	variable docs
<code>//@param name ...</code>	<code>@1</code>	F	param docs
<code>//@field name ...</code>	<code>@1</code>	T	field docs
<code>//# region / endregion</code>	<code>#</code>	—	editor regions
other <code>//</code>	<code>//</code>	—	ordinary comment

`helper._add_annotations` filters by suffix when attaching to nodes.

Store context

`_set_store_ctx` walks assignment targets (`Name` , nested `Tuple`) and sets `ctx = Store()` so load/store distinction is available to later passes (Python-ast style).



Representative visit methods

Rule visitor	Emits
<code>visitStart_script</code>	<code>Script(body)</code>
<code>visitStart_expression</code>	<code>Expression(body)</code>
<code>visitFunction_declaration</code> / <code>visitMethod_declaration</code>	<code>FunctionDef</code> (<code>method=1</code> for methods)
<code>visitType_declaration</code>	<code>TypeDef</code>
<code>visitEnum_declaration</code>	<code>EnumDef</code>
<code>visitCompound_*</code> / <code>visitSimple_*</code> assignment family	<code>Assign</code> / <code>ReAssign</code> / <code>AugAssign</code>
<code>visitIf_structure</code> / <code>visitFor_*</code> / <code>visitWhile_*</code> / <code>visitSwitch_*</code>	Structure expressions
<code>visit*_*_expression</code> (disjunction → ... → primary)	Operator trees with correct associativity
<code>visitLiteral_*</code>	<code>Constant</code> (numbers via <code>ast.literal_eval</code> , strings, bools, colors)
<code>visitPrimary_expression_call</code>	<code>Call</code> + <code>Arg</code> list (positional/keyword)

Expression climbing follows the grammar's layered rules (`conditional` → `disjunction` → `conjunction` → `equality` → `inequality` → `additive` → `multiplicative` → `unary` → `primary`).

Internals

Path

- Implementation: `src/pynescrypt/ast/builder.py`
- Visitor base: `src/pynescrypt/ast/grammar/antlr4/visitor.py` → generated `PynescryptParserVisitor`
- Node types: `src/pynescrypt/ast/node.py`
- Annotation pairing: `src/pynescrypt/ast/collector.py` + `helper._add_annotations`

Statement list flattening

`visitStatements` flattens each statement visitor's result: some rules return a **list** (simple multi-statement lines joined by commas), so the body is always a flat `list[stmt]` .

Export flag

When `EXPORT` is present on library declarations or compound name initialization, the builder sets `export = 1` on the corresponding node (integer flag per ASDL `int?`).

Field / enum members

Field definitions become `Assign` -like statements inside `TypeDef.body` (with optional `VarIp` mode). Enum members become assignment-shaped stmts under `EnumDef.body` . The unparser special-cases method members of types when pretty-printing.

Invariants

1. **Every constructed stmt/expr that has a source span should receive** `_setLocations` so LSP diagnostics and `get_source_segment` work.
2. **Visitor method names must track generated rule names.** Renaming a parser rule without updating the builder is a silent `generic_visit` fallback (wrong or empty trees).
3. **Do not edit** `builder.py.bak`. Live builder is `builder.py` only (stale backups are forbidden under project policy).
4. **Grammar and builder co-evolve.** A resource-only grammar change that adds rules is incomplete until `visit_*` exists.

Worked examples

What `x = close + 1` becomes

Rough structure after build:

```
Assign(
    target=Name(id="x", ctx=Store()),
    value=BinOp(
        left=Name(id="close", ctx=Load()),
        op=Add(),
        right=Constant(value=1),
    ),
)
```

Export const initialization

Grammar: `EXPORT? variable_declaration EQUAL structure_expression`

Builder sets `assign.export = 1` when `ctx.EXPORT()` is present — required for library export of typed constants.

Debugging a missing visitor

```
from pynescrypt.ast.helper import parse, dump
print(dump(parse(your_snippet), indent=2))
```

If a subtree is `None` or a raw unexpected type, find the rule in `PinescriptParser.g4` and the matching `visitRule_name` in `builder.py`.



Failure modes

Symptom	Cause
<code>AttributeError: '...Context' object has no attribute '...'</code>	Full parser regen changed accessors; builder outdated
Locations all zero / missing	Forgot <code>_setLocations</code> on new visit method
Annotations not attached	Comment kind suffix wrong, or statement not yielded by <code>StatementCollector</code>
Tuples not assignable	Store context not propagated through <code>Tuple.elts</code>
Structure expression lost	Wrong rule branch between statement vs expression path

See also

- [Grammar \(ANTLR4\)](#)
- [ASDL schema](#)
- [Helper API](#)
- [Error model](#)

Helper API

parse, unparse, dump, walk, literal_eval, and location utilities — the public AST surface.

Helper API

`src/pynascript/ast/helper.py` is the stable library face of the language core. It orchestrates ANTLR, the builder, annotations, and the unparser behind functions deliberately similar to CPython's `ast` module.

Abstract

Call `parse` to obtain an ASDL tree; `dump` / `walk` / `iter_*` to inspect it; `unparse` to regenerate source; `literal_eval` for constant-folding-safe evaluation of literal expressions. Location helpers (`copy_location` , `fix_missing_locations` , `get_source_segment` , `increment_lineno`) support tooling that rewrites or reports on trees.

Everything higher in the stack (CLI, LSP, Pro API, workers) ultimately routes through this module for front-end work.

Conceptual model

```

flowchart TB
    subgraph public ["Public API"]
        parse
        unparse
        dump
        walk
        literal_eval
        locs["location helpers"]
    end

    subgraph private ["Private pipeline"]
        _pis["_parse_inputstream / _parse_filestream"]
        _parse
        LEX["Lexer + TokenStream"]
        PAR["Parser + ErrorListener"]
        BLD["ASTBuilder.visit"]
        COL["StatementCollector"]
        ANN["_add_annotations"]
    end

    parse --> _pis --> _parse
    _parse --> LEX --> PAR --> BLD
    BLD --> COL --> ANN
    unparse --> UP["NodeUnparser"]
    literal_eval --> NLE["NodeLiteralEvaluator"]
  
```



Interface surface

`parse(source, filename="<unknown>", mode="exec") -> AST`

Parameter	Meaning
<code>source</code>	Full script or expression text
<code>filename</code>	Error reporting name; resolved to absolute path if the file exists
<code>mode</code>	"exec" → <code>start_script</code> → <code>Script</code> ; "eval" → <code>start_expression</code> → <code>Expression</code>

Raises:

- `ValueError` — invalid mode
- `pynescrypt.ast.error.SyntaxError` — lexer/parser failure (via `PinescriptErrorListener`)

Side effects during exec mode:

1. Temporarily raises `sys.getrecursionlimit` to at least 5000.
2. Collects COMMENT tokens and attaches annotations to eligible statements.

`unparse(node) -> str`

Delegates to `NodeUnparser().visit(node)`. See [Unparser](#).

`dump(node, *, annotate_fields=True, include_attributes=False, indent=None) -> str`

Recursive pretty-print of node trees. With `indent` (int spaces or string), multi-line layout is used for non-trivial nodes.

`literal_eval(node_or_string, context=None, data_feed=None, data_provider=None) -> Any`

- Strings are `parse(..., mode="eval")` first.
- Unwraps `Expression.body`.
- Uses `NodeLiteralEvaluator` — **not** full script evaluation. Non-literal graphs raise.

Tree navigation

Function	Behavior
<code>iter_fields(node)</code>	Yields (name, value) for set <code>_fields</code>
<code>iter_child_nodes(node)</code>	Direct AST children (incl. list items)
<code>walk(node)</code>	BFS over the full subtree



Location utilities

Function	Behavior
<code>copy_location(new, old)</code>	Copy <code>lineno/col/end_*</code> when both define the attribute
<code>fix_missing_locations(node)</code>	Fill gaps from parent defaults starting at <code>(1, 0)</code>
<code>increment_lineno(node, n=1)</code>	Shift all line numbers
<code>get_source_segment(source, node, *, padded=False)</code>	Slice original source by node span; <code>None</code> if incomplete ends

`__all__` export list

`copy_location`, `dump`, `fix_missing_locations`, `get_source_segment`, `increment_lineno`, `iter_child_nodes`, `iter_fields`, `literal_eval`, `parse`, `unparse`, `walk`.

Internals

Path

`src/pynascript/ast/helper.py` — all public helpers above.

Related:

- `src/pynascript/ast/collector.py` — `StatementCollector` for annotation pairing
- `src/pynascript/ast/unparser.py` — unparse backend
- `src/pynascript/ast/evaluator` — `NodeLiteralEvaluator` (`literal_eval` only)

Annotation algorithm (`_add_annotations`)

1. Merge comments and statements; sort by `(lineno, col_offset)`.
2. Group consecutive comments vs statements.
3. Keep only kinds starting with `@`.
4. Script-level: first group members with kind ending in `S` → `script.annotations`.
5. Pair remaining comment groups with following statements; attach `F / T / V` filters to `FunctionDef` / `TypeDef` / `Assign`.

Comment **nodes** are not left in `Script.body`; only string annotation lists are stored on targets.

Stream helpers

Internal	Role
<code>_parse_inputstream</code>	<code>InputStream(source)</code> + <code>stream.name = filename</code>
<code>_parse_filestream</code>	<code>FileStream</code> for on-disk scripts
<code>_get_absolute_path</code>	Resolve existing paths; leave <code><unknown></code> alone

Invariants

1. **mode** is a closed set. Only `exec` and `eval`.
2. **Default listeners are removed.** Console ANTLR spam is replaced by raising project `SyntaxError`.



3. `dump` requires an `AST` instance. Non-nodes raise `TypeError`.
4. Round-trip tests should use `parse` + `unparse`, not raw builder access, so annotation behavior matches production.

Worked examples

Inspect a tree

```
from pynescrypt.ast.helper import parse, dump, walk

tree = parse("""
//@version=5
indicator("demo")
plot(close)
""")
print(dump(tree, indent=2))
print(sum(1 for _ in walk(tree)), "nodes")
```

Source segment

```
from pynescrypt.ast.helper import parse, get_source_segment

src = "a = 1\nb = a + 2\n"
tree = parse(src)
assign_b = tree.body[1]
print(get_source_segment(src, assign_b)) # "b = a + 2\n" or similar span
```

Literal evaluation

```
from pynescrypt.ast.helper import literal_eval

assert literal_eval("2 * 3 + 4") == 10
assert literal_eval("'hi'") == "hi"
```

Failure modes

Failure	Notes
Deeply nested ternaries hit recursion	Mitigated by temporary limit ≥ 5000 ; pathological depth can still fail
<code>get_source_segment</code> returns <code>None</code>	Missing <code>end_lineno</code> / <code>end_col_offset</code>
<code>literal_eval</code> raises	Non-literal AST (names, calls beyond allowed set)
Filename <code><unknown></code> in errors	Expected when parsing pure strings without a path
Annotations missing	Comment not <code>@...</code> form, or not immediately preceding eligible stmt

See also

- [Builder](#)



- Unparser
- Visitor / transformer
- Error model

Unparser

NodeUnparser: precedence-aware AST → Pine Script source regeneration.

Unparser

The unparser is the inverse of the builder: ASDL nodes become source text with correct operator parenthesization and Pine idioms (`=>`, `:=`, `var` / `varip`, `export`, method vs function).

Abstract

`src/pynscript/ast/unparser.py` implements `NodeUnparser(NodeVisitor)` and a `Precedence` enum. `helper.unparse(node)` constructs a fresh unparser and returns `"".join` of the internal buffer. Output is **semantically faithful**, not a pretty-printer of original trivia: whitespace is normalized; comments that never became annotations are not reconstructed; annotation strings that *were* attached are re-emitted. Round-trip tests (`parse → unparse → parse`) are the primary oracle for grammar/builder/unparser coherence.

Conceptual model

```
flowchart LR
    AST["AST tree"]
    V["visit / traverse"]
    PR["Precedence table"]
    BUF["_source list"]
    OUT["source string"]

    AST --> V
    PR --> V
    V --> BUF --> OUT
```

Parenthesization rule of thumb: if a child expression's stored precedence is **weaker** (higher binding needed) than the parent operator, wrap the child in `()`.

Interface surface

```
from pynscript.ast.helper import parse, unparse

src = """
//@version=5
f(x) => x * 2
plot(f(close))
"""

print(unparse(parse(src)))
```



Precedence (low → high binding)

Level	Operators / forms
TEST	ternary <code>?</code> <code>:</code> (weakest)
OR	<code>or</code>
AND	<code>and</code>
EQ	<code>==</code> <code>!=</code>
INEQ / CMP	<code><</code> <code>></code> <code><=</code> <code>>=</code>
ARITH	<code>+</code> <code>-</code>
TERM	<code>*</code> <code>/</code> <code>%</code>
FACTOR / NOT	unary <code>+</code> <code>-</code> <code>not</code>
ATOM	names, literals, calls, subscripts

`Precedence.next()` steps one level tighter for right-hand associativity control.

Buffer helpers (internal but useful when subclassing)

Method	Role
<code>write(*text)</code>	Append fragments
<code>fill(text="")</code>	Newline + indent + text
<code>block()</code>	Indent context manager
<code>delimit(start, end)</code>	Wrap sub-output
<code>require_parens(prec, node)</code>	Conditional parentheses
<code>items_view</code> / <code>interleave</code>	Comma-separated lists
<code>buffered()</code>	Capture substring generation



Node coverage (high level)

Node	Emission sketch
Script	annotations then body
FunctionDef	optional <code>export / method</code> , <code>name(args) => body</code> (inline single <code>Expr</code> or indented block)
TypeDef	<code>type Name</code> with fields then methods
EnumDef	<code>enum Name</code> body
Assign	annotations, <code>export?</code> , <code>var / varip?</code> , <code>type?</code> , <code>target = value</code>
ReAssign	<code>target := value</code>
AugAssign	<code>target op= value</code>
ForTo / ForIn / While / If / Switch	Pine control syntax; <code>else if</code> chain collapse
Import	<code>import ns/name/version as alias?</code>
BinOp / BoolOp / UnaryOp / Compare / Conditional	precedence-aware
Call / Attribute / Subscript / Name / Constant / Tuple	primaries

Internals

Path

- `src/pynescrypt/ast/unparser.py` — `Precedence`, `NodeUnparser`
- Entry: `helper.unparse` → `NodeUnparser().visit(node)`

Visit vs traverse

`visit` resets `_source` and calls `traverse`. `traverse` accepts lists (statement bodies) or single nodes, then dispatches via the visitor cache. This split lets statement lists and expression trees share code without double-buffering.

If / else if chains

`visit_If` detects `orelse == [Expr(If(...))]` and emits `else if` rather than nested `else + if` blocks — matching idiomatic Pine.

Type body ordering

`visit_TypeDef` partitions body into field statements vs `FunctionDef` with `method` set, emitting fields first then methods.

Constants

String constants use JSON-style quoting helpers where needed; colors and numbers re-serialize from their Python values. Exact original quote style (single vs double) is not guaranteed.

Invariants

1. **Precedence tables must stay aligned with the parser's expression layers.** A mismatch causes either redundant parens (benign) or wrong binding after re-parse (severe).
2. **Annotations are part of the round-trip surface** when attached to Script/FunctionDef/TypeDef/Assign.
3. **Unparser does not type-check.** Ill-typed but well-shaped trees still print.
4. **Indent unit is four spaces** (`" " * self._indent`).

Worked examples

Precedence

```
from pynescrypt.ast.helper import parse, unparse

# Builder produces BinOp(Add, left=1, right=BinOp(Mult, 2, 3)) or inverse depending on parse
print(unparse(parse("1 + 2 * 3", mode="eval")))
# Expect something that re-parses to the same value: e.g. "1 + 2 * 3"
```

Function forms

```
# Single-expression body → f(x) => expr
# Multi-statement body → f(x) =>\n    stmt\n    stmt
```

Reassignment vs declaration

```
// Assign with mode → var float x = 1.0
// ReAssign → x := 2.0
// AugAssign → x += 1
```

Failure modes

Symptom	Cause
Missing <code>visit_*</code> output	Falls through <code>generic_visit</code> which only walks children — may emit empty string for leaf-like unhandled nodes
Extra parentheses everywhere	Over-conservative precedence
Re-parse differs in binding	Under-parenthesized output
Lost comments	Only annotation strings are stored; plain <code>//</code> comments are dropped after parse
Method printed as function	<code>FunctionDef.method</code> flag not set by builder

See also

- [Helper API](#)
- [Builder](#)



- ASDL schema
- Visitor / transformer



Visitor & Transformer

NodeVisitor and NodeTransformer: read-only traversal vs in-place AST rewrite with caching dispatch.

Visitor & Transformer

Tree walks in PYNE follow the same dual pattern as CPython's `ast` module: a visitor for analysis, a transformer for structural rewrite.

Abstract

- **NodeVisitor** (`src/pynescrypt/ast/visitor.py`) — dispatch `visit_<ClassName>` , default `generic_visit` walks children via `iter_fields` . Method lookups are cached per visitor instance.
- **NodeTransformer** (`src/pynescrypt/ast/transformer.py`) — same dispatch, but `generic_visit` **replaces** list items and fields according to return values (`None` removes, list splices, AST replaces).

Specialized visitors elsewhere:

Class	Path	Role
<code>PinescriptASTBuilder</code>	<code>builder.py</code>	CST visitor (ANTLR), not <code>NodeVisitor</code>
<code>NodeUnparser</code>	<code>unparser.py</code>	Source emission
<code>StatementCollector</code>	<code>collector.py</code>	Yield statements for annotations
Evaluator classes	<code>evaluator/*.py</code>	Runtime interpretation



Conceptual model

```
flowchart TB
    N["AST node"]
    D["visit(node)"]
    CACHE["_visitor_cache"]
    M["visit_ClassName or generic_visit"]

    N --> D
    D --> CACHE
    CACHE --> M

    subgraph visitor ["NodeVisitor.generic_visit"]
        V1["for field in iter_fields"]
        V2["visit children"]
        V3["discard return values"]
    end

    subgraph transformer ["NodeTransformer.generic_visit"]
        T1["for field in iter_fields"]
        T2["visit child"]
        T3["None → drop"]
        T4["list → extend"]
        T5["AST → replace"]
    end

    M --> visitor
    M --> transformer
```

Interface surface

NodeVisitor

```
from pynescrypt.ast.visitor import NodeVisitor
from pynescrypt.ast.helper import parse

class CallCounter(NodeVisitor):
    def __init__(self):
        super().__init__()
        self.count = 0

    def visit_Call(self, node):
        self.count += 1
        self.generic_visit(node)

tree = parse("plot(ta.sma(close, 14))")
c = CallCounter()
c.visit(tree)
assert c.count >= 2
```

Method	Contract
visit(node)	Resolve visit_<type(node).__name__> with cache; fallback generic_visit
generic_visit(node)	Recurse into AST fields and lists of AST

Returns are whatever the override returns (analysis often returns `None`).



NodeTransformer

```
from pynescrypt.ast import node as ast
from pynescrypt.ast.transformer import NodeTransformer
from pynescrypt.ast.helper import parse, unparse

class RenameClose(NodeTransformer):
    def visit_Name(self, node: ast.Name):
        if node.id == "close":
            return ast.Name(id="open", ctx=node.ctx)
        return node

tree = parse("plot(close)")
tree = RenameClose().visit(tree)
assert "open" in unparse(tree)
```

Return value semantics when transforming a **list parent** (e.g. `body`):

Return	Effect
None	Remove this element
AST	Replace element
non-AST iterable	Splice multiple nodes in place of one
same node	Keep

For a **single AST field**, `None` deletes the attribute (`delattr`); otherwise `setattr` replaces.

StatementCollector

Generator-style visitor used only for annotation attachment:

- Yields `FunctionDef`, `TypeDef`, `EnumDef`, `Assign`, `ReAssign`, `AugAssign`, `Import`, `Expr`, `Break`, `Continue`.
- Descends into nested structures and into structure-valued assignments.
- Does **not** yield the structure nodes themselves as statements when they appear only as values — it yields their inner bodies.

```
from pynescrypt.ast.collector import StatementCollector
from pynescrypt.ast.helper import parse

stmts = list(StatementCollector().visit(parse("f() => 1\nx = 2")))
```



Internals

Paths

File	Symbols
<code>src/pynascript/ast/visitor.py</code>	<code>NodeVisitor</code>
<code>src/pynascript/ast/transformer.py</code>	<code>NodeTransformer</code>
<code>src/pynascript/ast/collector.py</code>	<code>StatementCollector</code> , <code>Structure</code> tuple
<code>src/pynascript/ast/helper.py</code>	<code>iter_fields</code> , <code>iter_child_nodes</code> , <code>walk</code>

Dispatch cache

```
node_class = type(node).__name__
visitor = cache.get(node_class) or getattr(self, "visit_" + node_class, generic_visit)
```

Subclass instances that dynamically add methods after first visit of a type will not see them unless the cache is cleared — prefer defining methods on the class body.

In-place lists

Transformers mutate list fields with `old_value[:] = new_values` rather than rebinding, so aliased references to the same list see updates.

Invariants

1. **Always call** `super().__init__()` on subclasses so `_visitor_cache` exists.
2. **Transformers should return the node** from overrides that use `generic_visit` (the default returns `node` after rewriting children).
3. **Do not assume identity stability** after a transformer pass if you replaced roots.
4. `walk` **is not a visitor** — it is a pure BFS helper without dispatch.



Worked examples

Strip all Expr statements that call plot

```
from pyonescript.ast import node as ast
from pyonescript.ast.transformer import NodeTransformer
from pyonescript.ast.helper import parse, unparse

class DropPlots(NodeTransformer):
    def visit_Expr(self, node):
        if isinstance(node.value, ast.Call):
            func = node.value.func
            if isinstance(func, ast.Name) and func.id == "plot":
                return None
            return self.generic_visit(node)

tree = DropPlots().visit(parse("x = 1\nplot(x)\n"))
print(unparse(tree))
```

Collect all string constants

```
from pyonescript.ast.visitor import NodeVisitor
from pyonescript.ast.helper import parse

class Strings(NodeVisitor):
    def __init__(self):
        super().__init__()
        self.found = []

    def visit_Constant(self, node):
        if isinstance(node.value, str):
            self.found.append(node.value)

s = Strings()
s.visit(parse('indicator("hello")'))
```

Failure modes

Failure	Cause
AttributeError: _visitor_cache	Forgot <code>super().__init__()</code>
Transformer silently no-ops	Override returns nothing (<code>None</code>) unintentionally → node deleted
Infinite recursion	<code>visit_X</code> calls <code>self.visit(node)</code> on same node without change
Partial rewrite	Overrode <code>visit_*</code> without <code>generic_visit</code> — children untouched
Collector misses nested assign	Structure not in <code>Structure</code> tuple and not assigned as value

See also

- [Helper API](#)
- [Unparser](#)



- Builder
- ASDL schema

Type System

Pine v6 type model: qualifiers, builtins, collections, UDTs, TypeRegistry, and MethodResolver.

Type System

PYNE's type layer models Pine's *qualified* types and user-defined types (UDTs) as Python objects usable by analysis and evaluation — not as a full independent typechecker IR (yet), but as the shared vocabulary for “what kind of value is this?”

Abstract

`src/pynescrypt/ast/type_system.py` defines:

- **Qualifiers** — `const`, `simple`, `series`, `input` (`TypeQualifier`)
- **Builtin kinds** — `int`, `float`, `bool`, `string`, `color`, `na`, `enum`
- **Composite types** — `array<T>`, `matrix<T>`, `map<K,V>`
- **UDTs** — `UserDefinedType` with `Field s` and `MethodSignature s`
- **Runtime instances** — `ObjectInstance` for UDT values
- **Registry / resolution** — `TypeRegistry`, `MethodResolver` (`.new`, `.copy`, user methods)

ASDL already has `Qualify` / `Specialize` expression forms and `type_qual` constructors; the type system module is the **semantic** counterpart used when evaluating UDT construction and when tools reason about types.

Conceptual model

```
classDiagram
class Type {
+name: str
+qualifier: TypeQualifier?
+is_compatible_with(other)
}
class BuiltinType {
+kind: BuiltinTypeKind
}
class ArrayType {
+element_type: Type
}
class MatrixType {
+element_type: Type
}
class MapType {
+key_type: Type
+value_type: Type
}
class UserDefinedType {
+fields: dict
+methods: dict
+is_exported: bool
}
Type <|-- BuiltinType
Type <|-- ArrayType
Type <|-- MatrixType
Type <|-- MapType
Type <|-- UserDefinedType
UserDefinedType --> Field
UserDefinedType --> MethodSignature
ObjectInstance --> UserDefinedType
TypeRegistry --> UserDefinedType
TypeRegistry --> BuiltinType
MethodResolver --> TypeRegistry
```

Interface surface

Qualifiers

```
from pynescrypt.ast.type_system import TypeQualifier

TypeQualifier.CONST    # compile-time constant
TypeQualifier.SIMPLE   # bar-invariant
TypeQualifier.SERIES   # per-bar series
TypeQualifier.INPUT    # input.* style parameter
```

`Type.__str__` renders "series float" when a qualifier is set.



Builtins and factories

```
from pynescrypt.ast.type_system import (
    BuiltinTypeKind,
    int_type,
    float_type,
    bool_type,
    string_type,
    color_type,
    TypeQualifier,
)

t = float_type(TypeQualifier.SERIES) # "series float"
```

Factory	Kind
<code>int_type</code>	<code>INT</code>
<code>float_type</code>	<code>FLOAT</code>
<code>bool_type</code>	<code>BOOL</code>
<code>string_type</code>	<code>STRING</code>
<code>color_type</code>	<code>COLOR</code>

Collections

```
from pynescrypt.ast.type_system import ArrayType, MatrixType, MapType, int_type, string_type

ArrayType(int_type())           # array<int>
MatrixType(float_type())        # matrix<float>
MapType(string_type(), int_type()) # map<string, int>
```

UDTs

```
from pynescrypt.ast.type_system import UserDefinedType, Field, MethodSignature, float_type, int_type

point = UserDefinedType("point")
point.add_field(Field("x", float_type(), default_value=0.0))
point.add_field(Field("y", float_type(), default_value=0.0, varip=False))
point.add_method(MethodSignature("length", [], return_type=float_type()))
```

`Field` supports optional `default_value` and `varip` (mirrors `varip` fields in type declarations).

TypeRegistry

```
from pynescrypt.ast.type_system import TypeRegistry

reg = TypeRegistry()
reg.register_type(point)
assert reg.is_builtin_type("float")
assert reg.is_user_defined_type("point")
assert reg.get_type("float") is not None
```



Builtins are seeded in `_init_builtin_types` (`int`, `float`, `bool`, `string`, `color`, `na`, `enum`).

`ObjectInstance` + `MethodResolver`

```
from pynescrypt.ast.type_system import ObjectInstance, MethodResolver

inst = ObjectInstance(point)
inst.set_field("x", 1.5)
resolver = MethodResolver(reg)
# .new / .copy handled specially
copy = resolver.resolve_method(inst, "copy", [])
```

Method name	Behavior
<code>new</code>	Construct instance; positional args fill fields in declaration order
<code>copy</code>	Shallow copy of field dict
other	Lookup <code>UserDefinedType.methods</code> ; missing → <code>AttributeError</code>

Unknown field get/set on `ObjectInstance` raises `AttributeError` with type name context.

Relation to ASDL `type_qual`

ASDL	TypeQualifier
<code>Const</code>	<code>CONST</code>
<code>Input</code>	<code>INPUT</code>
<code>Simple</code>	<code>SIMPLE</code>
<code>Series</code>	<code>SERIES</code>

Builder produces `Qualify(qualifier=..., value=...)` for annotated types in the AST; the type system module is used when those need runtime/type-registry meaning (especially UDTs in builtins/evaluator).

Internals

Path

`src/pynescrypt/ast/type_system.py` — sole definition site for the classes above.

Consumers (outside this doc's page set, for orientation):

- Evaluator UDT / collection code (`evaluator/builtins/`, `matrix_evaluator`, etc.)
- Future/static analysis hooks (LSP hover may grow toward this model)

Compatibility

`Type.is_compatible_with` is currently shallow: `BuiltinType` compares by equality; the base method returns `False` for other pairs. Treat richer subtyping (series/simple lattice, numeric promotion) as **not fully encoded** here — bar-loop semantics still own much of the coercion behavior.



na and enum

Registered as builtin kinds for name lookup. Enum *definitions* in scripts appear as AST `EnumDef`; linking enum members into `BuiltinTypeKind.ENUM` instances is evaluator-side work.

Invariants

1. **Registry lookup prefers builtins** over UDTs of the same name.
2. **Field presence is schema-true**. Setting an undeclared field is an error (no open bags).
3. `.new` **does not type-check arguments** against field types in the resolver — it assigns positionally.
4. `varip` is metadata on `Field`, not a separate type qualifier enum member.

Worked examples

Model a simple UDT from Pine

```
type candle
  float o
  float h
  float l
  float c
```

```
from pynescrypt.ast.type_system import UserDefinedType, Field, float_type, TypeRegistry, MethodResolver

candle = UserDefinedType("candle")
for name in "o h l c".split():
    candle.add_field(Field(name, float_type()))
reg = TypeRegistry()
reg.register_type(candle)
c = MethodResolver(reg).resolve_method(ObjectInstance(candle), "new", [1, 2, 0.5, 1.5])
assert c.get_field("h") == 2
```

Qualified series float

```
from pynescrypt.ast.type_system import float_type, TypeQualifier
assert str(float_type(TypeQualifier.SERIES)) == "series float"
```

Failure modes

Failure	Cause
<code>AttributeError: Field '...' not found</code>	Typo or outdated UDT schema
<code>AttributeError: Method '...' not found</code>	Method never <code>add_method</code> 'd; not <code>new</code> / <code>copy</code>
Silent type confusion	<code>is_compatible_with</code> too weak — do not rely on it alone for safety
Name shadowing	UDT registered with a builtin name is unreachable via <code>get_type</code>



See also

- [ASDL schema](#) — `Qualify`, `TypeDef`, `EnumDef`
- [Runtime](#) — bar-loop use of values
- [Builder](#) — parsing type declarations
- [Linter](#) — style-level checks (not full typing)

Linter

PineLinter static rules: syntax via parse, version, deprecations, naming, style codes.

Linter

The core linter is a lightweight static checker that sits on the parse pipeline plus regex rules. It is not a full dataflow analyzer; it is the first automated gate for “does this even look like modern Pine?”

Abstract

`src/pynescrypt/ast/linter.py` exports:

- `LintWarning` — dataclass (code, message, line, column, severity)
- `PineLinter` — stateful runner accumulating warnings
- `lint_script(source, filename)` / `lint_file(filepath)` — convenience entry points

Rules run in fixed order: **syntax** → **version** → **deprecated patterns** → **naming** → **style**. Syntax failures become severity `"error"` with code `E001`; other rules are mostly `"warning"`.

Conceptual model

```

flowchart TB
    SRC["source string"]
    L["PineLinter.lint"]
    S["_check_syntax → parse()"]
    V["_check_version"]
    D["_check_deprecated"]
    N["_check_naming"]
    STY["_check_style"]
    OUT["list of LintWarning"]

    SRC --> L --> S --> V --> D --> N --> STY --> OUT
  
```

Interface surface

```

from pynescrypt.ast.linter import lint_script, PineLinter, LintWarning

warnings = lint_script("""
indicator("x")
plot(close)
""")
for w in warnings:
    print(w) # warning: [W001] Missing @version ... at line 1
  
```



LintWarning

Field	Type	Meaning
code	str	Stable id (E001 , W001 , C002 , ...)
message	str	Human-readable explanation
line	int None	1-based when known
column	int None	Reserved / optional
severity	str	"warning" (default) or "error"

`__str__` format: `{severity}: [{code}] {message} at {location}`.

`PineLinter.lint(source, filename=<input>) -> list[LintWarning]`

Resets `self.warnings`, runs all checks, returns the list (also stored on the instance).

File helper

```
from pynescrypt.ast.linter import lint_file

issues = lint_file("strategies/mean_reversion.pine")
```

Reads UTF-8 text then delegates to `lint_script`.

Rule catalog

Syntax

Code	Severity	Condition
E001	error	<code>parse(source, filename)</code> raises any exception; message includes exception text

Does not attempt recovery; one syntax error check per run.

Version

Code	Condition
W001	No <code>//@version = N</code> (flexible whitespace) match
W002	Version integer <code>< 5</code> (deprecated; suggest v5/v6)

Pattern: `//\s*@version\s*=\s*(\d+)`.

Deprecated patterns (`_check_deprecated`)

Code	Pattern (simplified)	Advice
W101	<code>security('EXCHANGE:SYM'...)</code> style	Prefer <code>request.security()</code> with explicit params
W102	<code>plot(... style=plot.style_histogram)</code>	Consider <code>plotcandle</code>
W103	<code>var int name = na</code>	Prefer <code>0</code> for type safety



Matches set `line` from prefix newline count. Case-insensitive search.

Naming (`_check_naming`)

Code	Condition
C001	LHS of <code>name = ta...</code> starts with lowercase — message suggests camelCase via <code>_to_camel</code>

Heuristic only: line-local regex `(\w+)\s*=\s*ta\.`

Style (`_check_style`)

Code	Condition
C002	Line length (<code>rstrip</code>) > 120
C003	Line matches <code>^\s+if\s+</code> — flagged as “single-line if without braces” heuristic
C004	File does not end with newline

Internals

Path

`src/pynescrypt/ast/linter.py`

Dependencies:

- `pynescrypt.ast.parse` (re-exported path via `from pynescrypt.ast import parse`)
- Standard library `re`, `dataclasses`

Design stance

The linter intentionally uses **regex over AST** for several rules so it still partially works when parse fails (version/deprecations/style still run after a failed syntax check — note: `_check_syntax` records `E001` but does not abort the pipeline). That means:

- False positives/negatives on fancy formatting are possible.
- Deep semantic issues (wrong series type, undefined names) belong to evaluator/LSP diagnostics, not these codes.

Integration points

CLI, editor save hooks, and CI can call `lint_script` without spinning up the full evaluator. LSP diagnostics may layer additional semantic checks beyond this module.

Invariants

- `lint()` always clears prior warnings on that instance before running.
- Codes are stable public strings — treat renames as breaking for tooling.
- Syntax errors are not fatal to the function — you get `E001` plus any later regex hits.
- No mutation of source — pure analysis.



Worked examples

Clean modern script

```
from pynescrypt.ast.linter import lint_script

src = """//@version=5
indicator("ok")
length = 14
basis = ta.sma(close, length)
plot(basis)
"""
assert lint_script(src) == [] # may still flag C001 if naming heuristic fires
```

Note: `basis = ta.sma(...)` triggers `C001` under the current rule (lowercase LHS). Prefer documenting that heuristic when teaching style.

Missing version

```
ws = lint_script('indicator("x")\nplot(close)\n')
assert any(w.code == "W001" for w in ws)
```

Programmatic filter

```
errors = [w for w in lint_script(src) if w.severity == "error"]
if errors:
    raise SystemExit(1)
```

Failure modes

Issue	Explanation
<code>C003</code> on legitimate indented <code>if</code> blocks	Regex is coarse; multi-line <code>if</code> with body still matches <code>^\s+if\s+</code>
<code>C001</code> noise	Many valid snake_case or lower identifiers
<code>E001</code> opaque message	Exception string from parse; see Error model for caret formatting when catching <code>SyntaxError</code> directly
Encoding errors in <code>lint_file</code>	Non-UTF-8 files raise at read time
False security deprecation	Pattern looks for quoted <code>EXCHANGE:SYM</code> form only

See also

- [Helper API](#) — underlying `parse`
- [Error model](#)
- [LSP diagnostics](#)
- [End-user troubleshooting](#)

Error Model

SyntaxError, IndentationError, SyntaxErrorDetails, and PinescriptErrorListener bridge from ANTLR to diagnostics.

Error Model

Parse failures in PYNE are not bare ANTLR console messages. They are structured exceptions with file, line, column, source excerpt, and a caret — suitable for CLI, tests, and LSP adapters.

Abstract

Two layers cooperate:

1. `src/pynscript/ast/error.py` — `SyntaxErrorDetails`, `SyntaxError`, `IndentationError`.
2. `src/pynscript/ast/grammar/antlr4/error_listener.py` — `PinescriptErrorListener` implements ANTLR's `ErrorListener.syntaxError`, builds details from the recognizer + offending token, and **raises** the project `SyntaxError`.

`helper._parse` installs this singleton listener on both lexer and parser after removing default listeners.

Indentation problems raised from `PinescriptLexerBase` use the same exception hierarchy (`IndentationError` subclass).

Note: these names intentionally shadow Python builtins (`SyntaxError`, `IndentationError`) inside the package — import carefully:

```
from pynscript.ast.error import SyntaxError as PineSyntaxError
```

Conceptual model

```
sequenceDiagram
    participant U as User code / parse()
    participant L as Lexer / Parser
    participant EL as PinescriptErrorListener
    participant E as SyntaxError

    U->>L: tokenize / parse
    L->>EL: syntaxError(recog, symbol, line, col, msg, exc)
    EL->>EL: filename, line text, end span from token
    EL->>E: raise SyntaxError(msg, details)
    E-->>U: exception with caret __str__
```



Interface surface

SyntaxErrorDetails (NamedTuple)

Field	Type	Meaning
filename	str	Path or <unknown>
lineno	int	1-based line
offset	int	0-based column
text	str	Source line (or excerpt)
end_lineno	int None	Multi-line span end
end_offset	int None	End column

SyntaxError(Exception)

```
class SyntaxError(Exception):
    def __init__(self, message: str, *details): ...
```

details may be:

- a single SyntaxErrorDetails instance, or
- positional components matching the NamedTuple fields.

Attributes: message, details.

__str__ renders:

```
{message}
File "{filename}", line {lineno}
  {stripped_line}
  {spaces}^
```

Offset is adjusted when the displayed line is `rstrip()`'d so the caret still points at the logical column.

IndentationError(SyntaxError)

Empty subclass for indent-stack failures from the lexer base (wrong nest, inconsistent levels). Catch either specifically or via the base `SyntaxError`.

PinescriptErrorListener

```
from pynescrypt.ast.grammar.antlr4.error_listener import PinescriptErrorListener

listener = PinescriptErrorListener.INSTANCE # singleton
lexer.removeErrorListeners()
lexer.addErrorListener(listener)
parser.removeErrorListeners()
parser.addErrorListener(listener)
```



Method	Role
<code>syntaxError(...)</code>	Build details; raise <code>SyntaxError</code>
<code>_getFilenameFrom(recognizer)</code>	Walk Parser → TokenStream → Lexer → InputStream/FileStream
<code>_getInputTextFrom(recognizer, lineno)</code>	Full buffer or single line
<code>_splitLines</code>	Portable newline split (<code>\r\n</code> / <code>\n</code> / <code>\r</code>)

If the ANTLR exception `e` is already a project `SyntaxError`, its `details` are updated; otherwise the new error's `__cause__` is set to `e`.

Internals

Paths

Path	Role
<code>src/pynescrypt/ast/error.py</code>	Exception types
<code>src/pynescrypt/ast/grammar/antlr4/error_listener.py</code>	ANTLR bridge
<code>src/pynescrypt/ast/grammar/antlr4/resource/PinescriptLexerBase.py</code>	May raise <code>IndentationError</code> / <code>SyntaxError</code> during tokenization
<code>src/pynescrypt/ast/helper.py</code>	Wires listener; maps filename onto streams

Filename resolution order (InputStream)

1. `getSourceName()` if present
2. `sourceName` attribute
3. `input_stream.name` (set by helper to the user filename)
4. `"<unknown>"`

`FileStream` uses `fileName`.

End span from offending token

```
symbol_len = stop - start + 1
end_lineno = token.line + newlines_in_text
end_offset = adjusted if multiline else column + symbol_len
```

Mirrors the builder's location math so diagnostics and AST spans speak the same coordinate system.

Relationship to linter `E001`

`PineLinter._check_syntax` catches **any** `Exception` from `parse` and wraps it as `LintWarning(code="E001", severity="error")`, losing structured details. For rich carets, catch `pynescrypt.ast.error.SyntaxError` at the call site instead of only reading linter output.

LSP / tooling

Adapters typically map:



Exception field	LSP Diagnostic
<code>lineno</code> / <code>offset</code>	<code>Range.start</code>
<code>end_lineno</code> / <code>end_offset</code>	<code>Range.end</code>
<code>message</code>	<code>message</code>
<code>IndentationError</code>	same, possibly different <code>code</code>

Invariants

1. **Default ANTLR listeners must stay removed** on the parse path — otherwise errors print twice and may not raise.
2. **Singleton listener is stateless** enough to share (`INSTANCE`); thread-local needs would require new instances.
3. **Raising aborts parse immediately** — no error recovery producing a partial tree through the public helper.
4. **Package `SyntaxError` ≠ builtin** — `except SyntaxError` without import may catch the wrong class.

Worked examples

Catching a parse error

```
from pynescrypt.ast.helper import parse
from pynescrypt.ast.error import SyntaxError as PineSyntaxError

try:
    parse("f( =>\n", filename="bad.pine")
except PineSyntaxError as e:
    print(e) # caret form
    print(e.details.lineno, e.details.offset)
    print(e.details.filename)
```

Distinguishing indentation

```
from pynescrypt.ast.error import IndentationError as PineIndentError
from pynescrypt.ast.error import SyntaxError as PineSyntaxError

try:
    parse("if true\nplot(1)\n") # missing indent under if – may indent-error depending on tokens
except PineIndentError as e:
    print("indent", e.message)
except PineSyntaxError as e:
    print("syntax", e.message)
```



Manual construction (tests)

```
from pyonescript.ast.error import SyntaxError, SyntaxErrorDetails

details = SyntaxErrorDetails("t.pine", 3, 4, "    plot(\n", 3, 9)
err = SyntaxError("missing RPAR", details)
assert "t.pine" in str(err)
assert "^" in str(err)
```

Failure modes

Failure	Cause
<code>TypeError: unexpected type of input</code>	Listener received a recognizer/stream combo it does not support
Caret misaligned	Tabs vs spaces; display strips leading WS while offset counts raw
Filename always <code><unknown></code>	Stream name not set (bypassed helper)
Exception swallowed as generic <code>E001</code>	Only used linter path
<code>except SyntaxError</code> misses Pine errors	Caught builtin only
Partial trees	Not available via <code>parse()</code> after error — redesign would need recovery listeners

See also

- [Helper API](#)
- [Grammar \(ANTLR4\)](#)
- [Linter](#)
- [LSP diagnostics](#)
- [Language core hub](#)